

Task-2

Student Management System – Deliverables

Submitted Person Details	
Name	Mude Meghana
Register No	23691A05A3
College	Madanapalle Institute of Technology and Science
Task	02 (JAVA DEVELOPER)
Submitted Date	07/06/2026
E-mail	meghanamude@gmail.com

1. Java Source Code Files

Create Student class:

```
package com.maincrafts.server1;
```

```
public class Student {
```

```
    private int id;
```

```
    private String name;
```

```
    private int age;
```

```
    private String dob;
```

```
    private String course;
```

```
    private String collegeName;
```

```
    public Student(int id, String name, int age, String dob, String course, String collegeName) {
```

```
        super();
```

```
        this.id = id;
```

```
        this.name = name;
```

```
        this.age = age;
```

```
this.dob = dob;

this.course = course;

this.collegeName = collegeName;
}

public static Student fromString(String line) {

    String[] data = line.split(",");

    return new Student(
        Integer.parseInt(data[0]),
        data[1],
        Integer.parseInt(data[2]),
        data[3],
        data[4],
        data[5]
    );
}

public int getId() {
    return id;
}

public void setId(int id) {
    this.id=id;
}

public String getName() {
    return name;
}
```

```
public void setName(String name) {
```

```
    this.name = name;
```

```
}
```

```
public int getAge() {
```

```
    return age;
```

```
}
```

```
public void setAge(int age) {
```

```
    this.age = age;
```

```
}
```

```
public String getDob() {
```

```
    return dob;
```

```
}
```

```
public void setDob(String dob) {
```

```
    this.dob = dob;
```

```
}
```

```
public String getCourse() {
```

```
    return course;
```

```
}
```

```
public String getCollegeName() {
```

```
    return collegeName;
```

```
}
```

```
public void setCollegeName(String collegeName) {
```

```
    this.collegeName = collegeName;
```

```
}
```

```

public void setCourse(String course) {
    this.course = course;
}

@Override

public String toString() {
    return "Student [id=" + id + ", " + (name != null ? "name=" + name + ", " : "") + "age=" + age
    + ", "
    + (dob != null ? "dob=" + dob + ", " : "") + (course != null ? "course=" + course + ", " : "")
    + (collegeName != null ? "collegeName=" + collegeName : "") + "]";
}
}

```

Create a file manager to handle student class:

```

package com.maincrafts.server1;

import java.io.BufferedReader;

import java.io.FileReader;

import java.io.FileWriter;

import java.io.IOException;

import java.util.*;

public class Filehandling {

    public static final String filename="Student.txt";

    public void SaveFile(Student student)
    {
        try(FileWriter fw=new FileWriter(filename,true))
        {
            fw.write(student.toString() + "\n");
        }
        catch(IOException e)

```

```

{
    System.out.println("Error! we cannot save the students data");
}
}

public static List<Student> loadStudents()
{
    List<Student> students=new ArrayList<>();

    try(BufferedReader br=new BufferedReader(new FileReader(filename)))
    {
        String line;

        while((line=br.readLine()) != null)
        {
            students.add(Student.fromString(line));
        }
    }

    catch(Exception e)
    {
        System.out.println(e.getMessage());
    }

    return students;
}
}

```

Create a main function file to run those files:

```
package com.maincrafts.server1;
```

```
import java.util.List;
```

```
import java.util.Scanner;
```

```
public class StudentManagement {
```

```

private static List<Student> students;

public static void main(String[] args) {

    // TODO Auto-generated method stub

    students=Filehandling.loadStudents();

    Scanner sc=new Scanner(System.in);

    //int choice=sc.nextInt();

    while(true)
    {

        System.out.println("\n=====Student Management
System=====");

        System.out.println("1.Add Students");
        System.out.println("2.View Students");
        System.out.println("3.Search Students");
        System.out.println("4.Exit");
        System.out.println("Enter Your choice:");
        int choice=sc.nextInt();
        switch(choice)
        {
            case 1->AddStudents(sc);
            case 2->ViewStudents();
            case 3->SearchStudents(sc);
            case 4 ->
            {
                sc.close();

                System.out.println("ThankYou For visiting this file.");

                System.exit(0);

            }

            default -> System.out.println("Invalid Choice");

```

```

    }
    }
}

private static void ViewStudents() {
    // TODO Auto-generated method stub
    if(students.isEmpty())
    {
        System.out.println("No students Found");
        return;
    }
    for(Student s:students)
    {
        System.out.println(s);
    }
}

private static void SearchStudents(Scanner sc) {
    // TODO Auto-generated method stub
    System.out.print("Enter id:");
    int id=sc.nextInt();
    for(Student s:students) {
        if(s.getId()==id)
        {
            System.out.println("Students details found");
            System.out.println(s);
            return;
        }
    }
}

```

```

        System.out.println("Not Found");

    }

    private static void AddStudents(Scanner sc) {

        // TODO Auto-generated method stub

        System.out.print("Enter id:");

        int id=sc.nextInt();

        sc.nextLine();

        System.out.print("Enter name:");

        String name=sc.nextLine();

        System.out.print("Enter age:");

        int age=sc.nextInt();

        sc.nextLine();

        System.out.print("Enter DOB:");

        String dob=sc.nextLine();

        System.out.print("Enter course:");

        String course=sc.nextLine();

        System.out.print("Enter College Name:");

        String collegeName=sc.nextLine();

        Student student=new Student(id,name,age,dob,course,collegeName);

        students.add(student);

        Filehandling filehandling = new Filehandling();

        filehandling.SaveFile(student);

        System.out.println("data stored Successfully!");

    }

}

```


Output:

For input string: "Student [id=101"

=====Student Management System=====

1.Add Students

2.View Students

3.Search Students

4.Exit

Enter Your choice:

1

Enter id:1

Enter name:megha

Enter age:20

Enter DOB:07-may-2006

Enter course:CSE

Enter College Name:mits

data stored Successfully!

=====Student Management System=====

1.Add Students

2.View Students

3.Search Students

4.Exit

Enter Your choice:

1

Enter id:2

Enter name:Hemanth

Enter age:23

Enter DOB:10-sep-2004

Enter course:CSE

Enter College Name:MBU

data stored Successfully!

=====Student Management System=====

1.Add Students

2.View Students

3.Search Students

4.Exit

Enter Your choice:

1

Enter id:3

Enter name:Manoj

Enter age:18

Enter DOB:CST

Enter course:

Enter College Name:

data stored Successfully!

=====Student Management System=====

1.Add Students

2.View Students

3.Search Students

4.Exit

Enter Your choice:

2

Student [id=1, name=megha, age=20, dob=07-may-2006, course=CSE,
collegeName=mits]

Student [id=2, name=Hemanth, age=23, dob=10-sep-2004, course=CSE,
collegeName=MBU]

Student [id=3, name=Manoj, age=18, dob=16-April-2008, course=CST,
collegeName=MIT]

=====Student Management System=====

1.Add Students

2.View Students

3.Search Students

4.Exit

Enter Your choice:

3

Enter id:1

Students details found

Student [id=1, name=megha, age=20, dob=07-may-2006, course=CSE,
collegeName=mits]

=====Student Management System=====

1.Add Students

2.View Students

3.Search Students

4.Exit

Enter Your choice:

4

ThankYou For visiting this file.